Set-Based Robust Control Co-Design: Application to a Hybrid Thermal Management System

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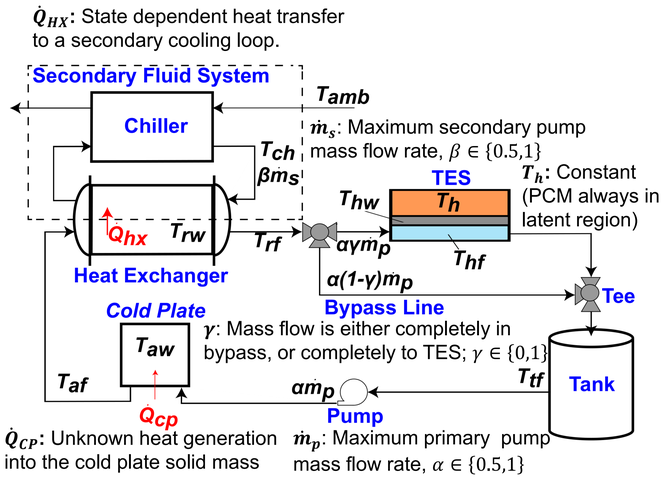
Motivation

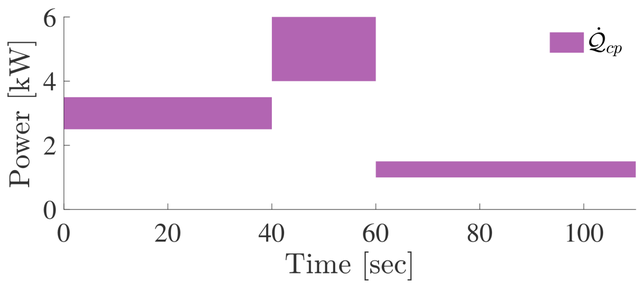
- **Thermal management systems** (TMSs) integrated with phase-change **thermal energy storage** (TES) devices provide robustness against highly transient heat loads produced by electrical systems
- TES provide additional heat rejection capacity only when needed, so its operation must be **actively controlled**
- **Control co-design** can be used to effectively design the system and controller and guarantee safe system operation.

System Description

The **single-phase cooling loop** considered in this work is shown below. The cold plate, heat exchanger, and thermal energy storage (TES) are each modeled as a **lumped parameter system**. In addition, the tank is modeled as a single state, assuming perfect mixing.



Problem Statement



Design disturbances set $\mathcal{V}$

The **heuristic controller** is formulated to determine the values of α, β and γ, which correspond to the primary pump speed, secondary pump speed, and whether the TES is being utilized.

Table 1. Definition of admissible system modes

Mode	α	β	γ	Description
q_1	0.5	0.5	0	Bypass, Low $\dot{m}$
q_2	0.5	0.5	1	TES, Low $\dot{m}$
q_3	1	1	0	Bypass, High $\dot{m}$
q_4	1	1	1	TES, High $\dot{m}$

Every t_D seconds, the discrete state σ_k is updated according to a set of IF-THEN-ELSE rules:

IF g_LTHEN $q_L = q_2$ ELSE $q_L = q_1$
IF g_HTHEN $q_H = q_4$ ELSE $q_L = q_3$
IF g_MTHEN $\sigma_k = q_H$ ELSE $\sigma_k = q_L$

$g_L = True$ when TES should be used, assuming low $\dot{m}$ is appropriate
$g_H = True$ when TES should be used, assuming high $\dot{m}$ is appropriate
$g_M = True$ when high $\dot{m}$ is appropriate

We aim to design a safe closed loop system using **set-based robust control co-design (RCCD)**.
Set-based: Safety verification is conducted through set operations
Robust: The system is safe for all time when subject to a periodic disturbance set $\mathcal{V}$.
CCD: Simultaneous design of system and controller

Set Definitions:
$\mathcal{R}_0$: Set of all possible initial conditions
$\mathcal{V} = \{\mathcal{V}_0 \cdots \mathcal{V}_{N-1}\}$: Time-varying disturbances set
$\mathcal{X} = \{\mathcal{X}_0 \cdots \mathcal{X}_N\}$: Time-varying safe set
$\mathcal{U}$: Set of all admissible control actions
$\mathcal{P}$: The set of all admissible parameters

Optimization Problem Formulation

Optimization Statement:
1. Optimize the size of a set of costs for the tube of all possible trajectories originating from the operating region $\mathcal{R}_0$.
2. Minimize the cost of design variables p while ensuring that p belongs to the admissible parameter set $\mathcal{P}$.

Furthermore, the closed-loop system must robustly remain within a safe subset $\mathcal{X}$ for all possible disturbances belonging to the time-varying set $\mathcal{V}$ for all time under a feedback control policy π that results in a sequence of control actions all belonging to $\mathcal{U}$.

$$\min_{\mathrm{p}} \; m(N,p,\mathcal{R}_N) + \sum_{k=0}^{N-1} l(k,p,\mathcal{R}_k,\mathcal{V}_k)$$
$$s.t. \mathcal{R}_{k+1} = suc(\mathcal{R}_k,\mathcal{V}_k,p),$$
$$\mathcal{R}_{k+1} \subset \mathcal{X},$$
$$\mathcal{R}_N \subset \mathcal{R}_0,$$
$$\{\pi(x,p_c)|x \in \mathcal{R}_k\} \subset \mathcal{U},$$
$$p \in \mathcal{P},$$
$$\forall k \in \{0,\cdots,N-1\}.$$

Note: $p = \begin{bmatrix} p_p^T & p_c^T \end{bmatrix}^T$, where p_p are plant parameters and p_c are controller parameters.

Note: m is the mass of all control volumes in the system model

$$\min_{\mathrm{p}} \; m + w_1(\dot{m}_p + \dot{m}_s) - w_2\dot{Q}_{cp,M}$$
$$s.t. \mathcal{R}_{k+1} = suc(\mathcal{R}_k,\mathcal{V}_k,p),$$
$$\rho^+(\mathcal{R}_k,1) \le T_{aw,H}$$
$$\rho^+(\mathcal{R}_N,i) \le \rho^+(\mathcal{R}_0,i),$$
$$\rho^-(\mathcal{R}_0,8) \le \rho^-(\mathcal{R}_N,8),$$
$$\mathrm{p_{lb}} \le p \le p_{ub},$$
$$\forall i \in \{1,\cdots,7\}.$$
$$\forall k \in \{0,\cdots,N-1\}.$$

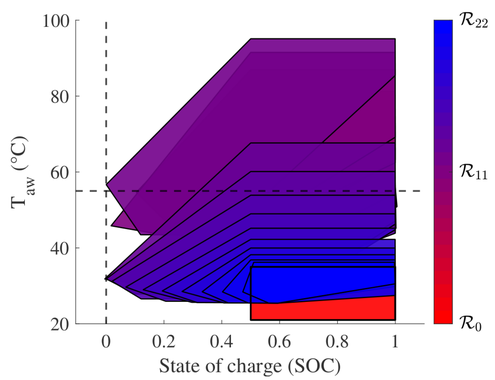
Definition: The support functions ρ^+ and ρ^- of a set $\mathcal{Z} \subset \mathbb{R}^n$ in the cardinal direction of the i^{th} dimension are given by

$$\rho^+(\mathcal{Z},i) = \sup\{z[i]\ z \in \mathcal{Z}\}$$
$$\rho^-(\mathcal{Z},i) = \inf\{z[i]\ z \in \mathcal{Z}\}$$

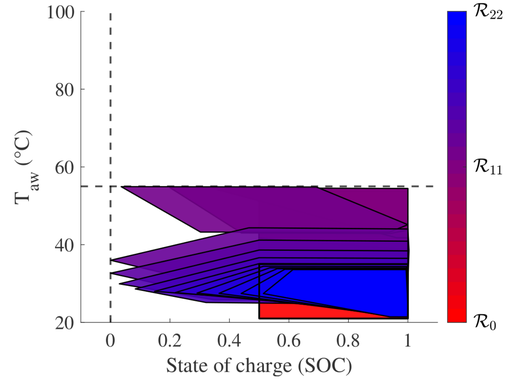
Optimization Results: Baseline and Set-Based

The solution to the presented optimization is compared to a baseline optimization, which checks safety only for maximum disturbance.

	Baseline	Set-Based
Solution Time	6 minutes	5.7 hour
Safe?	Not Safe	Safe



Above: Projection onto T_{aw} and SOC dimensions of the set of states reached at each time step using the **baseline** parameters.



Above: Projection onto T_{aw} and SOC dimensions of the set of states reached at each time step using **set-based** parameters.

Summary & Future Work

Key Contributions
- Applied set-based RCCD to a hybrid thermal management system.

Future Work
- Explore other methodologies to solve this problem that are more computationally efficient and with more favorable guarantees on finding the globally optimal solution.

Acknowledgements

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